

A COMPOSITE COEFFICIENT FOR QUANTIFYING ETHICAL ROBUSTNESS IN ADVERSARIAL MACHINE-LEARNING SYSTEMS

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ABSTRACT. Machine-learning systems deployed in scientific and high-stakes settings are routinely certified by a single scalar—accuracy—that is provably blind to the ethical failure modes (bias, integrity, transparency collapse) that adversarial prompts can induce while correctness is preserved. We formalize the informal hypothesis that models trained under explicit ethical constraints are more robust under adversarial perturbation, and that this gain is captured by a composite metric. We introduce the *composite ethical-robustness coefficient* $RC_\lambda(f, x, \varepsilon) = \lambda \rho_C + (1 - \lambda) \rho_A$, a convex combination of the worst-case preservation ratios of scientific correctness and ethical alignment over an adversarial perturbation set, and prove six results about it. We show that RC_λ is well-posed (bounded in $[0, 1]$, monotone non-increasing in the perturbation radius, exact at radius zero); that its deficit admits an exact TRADES-style additive decomposition into correctness- and alignment-stability terms; and—our central result—that a training-time constrained-Markov-decision-process alignment constraint, together with a bounded correctness cost, forces a provable separation between the constrained and the accuracy-only model that exceeds the target 0.15 on an explicit, checkable parameter region. We further prove that the composite is *necessary*: every correctness-only metric is provably blind to alignment collapse, whereas RC_λ separates a safe from a collapsed model by $1 - \lambda$. Finally, we give attack-independent Lipschitz and randomized-smoothing certified lower bounds. Every algebraic step is machine-checked with a computer-algebra system, and every quantitative claim with a seeded Monte-Carlo study.

1. INTRODUCTION

1.1. Motivation. Machine-learning systems in scientific and safety-critical domains are almost universally judged by a single scalar: accuracy, or one of its robust variants. This is convenient, but it is also mathematically incomplete. A model may answer a scientific question correctly and yet, under an adversarially crafted prompt, silently violate the ethical requirements we care about—fabricating citations, amplifying bias, leaking private information, or abandoning transparency—while its answer remains nominally right. An accuracy-only report certifies such a model as safe. The gap is not a matter of measurement noise: it is structural. A metric that depends on the model only through its correctness behavior *cannot*, as a matter of logic, detect a failure that leaves correctness untouched.

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This motivates a precise mathematical question. Adversarial-robustness theory has produced rigorous, quantitative coefficients for the stability of *correctness* under perturbation—the CLEVER score built from local Lipschitz constants [8], the certified radius of randomized smoothing [5], and the exact robust-error decomposition of TRADES [9]. Alignment research has, in parallel, produced quantitative measures of *ethical* stability, chiefly the attack success rate against standardized jailbreak benchmarks [4, 10]. These two literatures have not been fused: to our knowledge no prior work combines a certified-robustness-style correctness term with an alignment-stability term into a single coefficient with provable properties, and none connects a training-time ethical constraint to a lower bound on such a coefficient.

1.2. The question. We study the following informal hypothesis, which arises repeatedly in the empirical literature on aligned models: *models trained with explicit ethical constraints are at least 0.15 more robust than accuracy-only models under adversarial prompts, and this improvement is captured by a composite metric.* We ask whether this can be turned into a theorem. Concretely: is there a precisely defined coefficient RC_λ with provable analytic properties, together with explicit sufficient conditions under which a training-time ethical constraint *guarantees* a separation

$$\Delta = \mathbb{E}[\text{RC}_\lambda^{\text{eth}}] - \mathbb{E}[\text{RC}_\lambda^{\text{acc}}] \geq 0.15?$$

And is such a composite *necessary*, or does accuracy alone already suffice?

1.3. Contributions. We answer both questions affirmatively. We define the composite ethical-robustness coefficient

$$\text{RC}_\lambda(f, x, \varepsilon) = \lambda \rho_C(f, x, \varepsilon) + (1 - \lambda) \rho_A(f, x, \varepsilon), \quad \lambda \in [0, 1],$$

where $\rho_C, \rho_A \in [0, 1]$ are the worst-case preservation ratios of scientific correctness C and ethical alignment A over an adversarial perturbation set $B(x, \varepsilon)$, and prove:

- **Well-posedness (Theorem 3.1).** RC_λ is bounded in $[0, 1]$, non-increasing in ε , equal to 1 at $\varepsilon = 0$, and recovers the single-term correctness and alignment metrics at $\lambda = 1$ and $\lambda = 0$.
- **Exact decomposition (Theorem 3.3).** The deficit of each component obeys an exact additive identity $e_{\text{rob}}^S = e_{\text{nat}}^S + b^S$, the direct analogue of the TRADES identity $R_{\text{rob}} = R_{\text{nat}} + R_{\text{bdy}}$ [9], yielding a normalized upper bound on $1 - \mathbb{E}[\text{RC}_\lambda]$.
- **Constraint $\Rightarrow$ separation (Theorem 3.4, central).** A constrained-Markov-decision-process alignment constraint on the ethical model, together with a bounded correctness cost δ_C , forces $\Delta \geq (1 - \lambda)[(1 - d) - \bar{a}/a_{\min}] - \lambda\delta_C$, which meets the 0.15 target on an explicit region.
- **Necessity of the composite (Theorem 3.7).** There exist two models indistinguishable to every correctness-only metric yet separated by RC_λ ; hence $\lambda < 1$ is necessary to detect alignment collapse and $\lambda > 0$ to detect correctness collapse.
- **Certified lower bounds (Theorem 3.8).** RC_λ admits attack-independent Lipschitz and randomized-smoothing certified lower bounds.
- **Admissible-weight region (Theorem 3.10).** The set of weights λ achieving any target separation is characterized in closed form.

Although the correctness score C is specialized in our motivating application to correctness on astrophysics questions, every theorem below is stated for arbitrary measurable scores $C, A \in [0, 1]$ and is therefore domain-agnostic.

1.4. Techniques. The proofs are elementary but structured. Well-posedness and the decomposition follow from the nesting property of perturbation sets and linearity of expectation. The central separation theorem imports constrained-Markov-decision- process feasibility [1] as the mechanism that turns an ethical *training constraint* into an alignment-robustness *floor*, and combines it with a clean-competence floor via two elementary bounds on the preservation ratio. Necessity is proved by an explicit two-model hard-instance construction in the style of [7]. The certified bounds instantiate the Lipschitz margin argument of [8] and the Neyman–Pearson certificate of [5]. Every symbolic identity and inequality is verified with a computer-algebra system, and every quantitative claim is checked by a seeded Monte-Carlo study; Section 4 reports these.

1.5. Organization. Section 2 fixes notation, states the definitions, and recalls the prerequisite results. Section 3 states and proves the six theorems, with worked examples. Section 4 reports the computational verification and discusses implications, limitations, and open problems. Section 5 concludes.

2. PRELIMINARIES

2.1. Notation. Inputs x are drawn from a distribution D on an input space $\mathcal{X}$, and f denotes a fixed model. Expectations $\mathbb{E}[\cdot]$ are over $x \sim D$ unless otherwise indicated. We write Φ for the standard-normal cumulative distribution function and Φ^{-1} for its inverse. For $a \leq b$ and a real number t we write $\text{clip}(t, a, b) = \min\{\max\{t, a\}, b\}$.

2.2. Perturbation sets. An adversary is modeled by a family of *perturbation sets* $\{B(x, \varepsilon)\}_{\varepsilon \geq 0}$, one per input, indexed by a radius $\varepsilon \geq 0$. We require two structural axioms.

Definition 2.1 (Admissible perturbation family). A family $\{B(x, \varepsilon)\}$ is *admissible* if it satisfies

- (P0) $B(x, 0) = \{x\}$ for every x ;
- (P1) (*nesting*) $\varepsilon_1 \leq \varepsilon_2$ implies $B(x, \varepsilon_1) \subseteq B(x, \varepsilon_2)$.

Both axioms hold for ℓ_p balls $B_p(x, \varepsilon) = \{x' : \|x' - x\|_p \leq \varepsilon\}$ and, in the prompt setting, for families of adversarial prompts measured by suffix length, edit distance, or token-substitution radius; this is the sense in which our results specialize to the discrete adversarial-prompt threat models of [10, 4].

2.3. Scores and preservation ratios. The model is scored on two axes.

Definition 2.2 (Scores and worst-case scores). Let $C(f, x) \in [0, 1]$ and $A(f, x) \in [0, 1]$ be the clean *scientific-correctness* and *ethical-alignment* scores at x ; the alignment score aggregates dimensions such as bias, reproducibility, transparency, integrity, and privacy, and is of the form $A = 1 - \text{ASR}$ in the sense of [4]. The *worst-case scores* over $B(x, \varepsilon)$ are

$$\bar{C}(f, x, \varepsilon) = \min_{x' \in B(x, \varepsilon)} C(f, x'), \quad \bar{A}(f, x, \varepsilon) = \min_{x' \in B(x, \varepsilon)} A(f, x').$$

By (P0) and (P1), $\bar{C} \leq C$, $\bar{A} \leq A$ pointwise, with equality at $\varepsilon = 0$.

Definition 2.3 (Component robustness and composite coefficient). Fix a floor $\tau \in (0, 1)$. The *component robustness ratios* are

$$\rho_C(f, x, \varepsilon) = \text{clip}\left(\frac{\bar{C}(f, x, \varepsilon)}{\max(C(f, x, \tau))}, 0, 1\right), \quad \rho_A(f, x, \varepsilon) = \text{clip}\left(\frac{\bar{A}(f, x, \varepsilon)}{\max(A(f, x, \tau))}, 0, 1\right).$$

For a weight $\lambda \in [0, 1]$, the *composite ethical-robustness coefficient* is

$$\text{RC}_\lambda(f, x, \varepsilon) = \lambda \rho_C(f, x, \varepsilon) + (1 - \lambda) \rho_A(f, x, \varepsilon),$$

and its population version is $\text{RC}_\lambda(f, \varepsilon) = \mathbb{E}_x[\text{RC}_\lambda(f, x, \varepsilon)]$.

The floor τ guards against division by a vanishing clean score; it is active only on the event $\{C < \tau\}$ or $\{A < \tau\}$, and the outer clip ensures $\rho_C, \rho_A \in [0, 1]$ regardless.

Definition 2.4 (Deficit quantities). For a score $S \in \{C, A\}$ write the *natural deficit* $e_{\text{nat}}^S = \mathbb{E}[1 - S]$, the *robust deficit* $e_{\text{rob}}^S = \mathbb{E}[1 - \bar{S}]$, and the *stability gap* $b^S = \mathbb{E}[S - \bar{S}] \geq 0$.

2.4. Regularity and constraint hypotheses. Several results use clean-competence floors and Lipschitz regularity.

Definition 2.5 (Clean floors and Lipschitz constants). We say the model has *clean floors* $c_{\min}, a_{\min} \in (0, 1]$ if $C(f, x) \geq c_{\min}$ and $A(f, x) \geq a_{\min}$ almost surely. We say $C(f, \cdot)$ is L_C -Lipschitz on $B(x, \varepsilon)$ if $|C(f, x') - C(f, x'')| \leq L_C d(x', x'')$ for all $x', x'' \in B(x, \varepsilon)$, and similarly L_A for $A(f, \cdot)$, where $d(\cdot, \cdot)$ is the metric defining the perturbation set.

The mechanism connecting ethical training to robustness is modeled after the constrained-Markov-decision-process (CMDP) feasibility that constrained policy optimization enforces [1]. Concretely, we distinguish two training regimes through their alignment behavior under the adversary.

Definition 2.6 (Constraint and collapse models). The *ethically constrained* model f_{eth} satisfies the alignment constraint

$$(EC) \quad \mathbb{E}[1 - \bar{A}(f_{\text{eth}}, \cdot, \varepsilon)] \leq d$$

for a tolerance $d \in [0, 1)$, equivalently $\mathbb{E}[\bar{A}_{\text{eth}}] \geq 1 - d$. The *accuracy-only* model f_{acc} may collapse under the adversary:

$$(CO) \quad \mathbb{E}[\bar{A}(f_{\text{acc}}, \cdot, \varepsilon)] \leq \bar{a}$$

for a collapse level $\bar{a} \in [0, 1]$.

Constraint equation EC is exactly the CMDP feasibility $J_C(\pi_{\text{eth}}) \leq d$ of [1, Def. 2, Prop. 2], realizable in practice by the constitution of Constitutional AI [3] or the KL-regularized reward of RLHF [6, 2]; the collapse equation CO models the empirically documented jailbreak vulnerability of unconstrained models [10].

2.5. Prerequisite results. We use, without reproof, the following classical statements.

Theorem 2.7 (TRADES decomposition [9, Def. 2.1–2.3]). *For a binary classifier the robust error decomposes exactly as $R_{\text{rob}}(f) = R_{\text{nat}}(f) + R_{\text{bdy}}(f)$, a clean-error term plus a non-negative boundary term.*

Theorem 2.8 (Certified radius [5, Thm. 1]). *Let g be the Gaussian smoothing of a base classifier with variance σ^2 . If at x the top and runner-up class probabilities under the smoothing noise satisfy $p_A \geq p_B$, then g is constant on the ℓ_2 -ball of radius $R = \frac{\sigma}{2}(\Phi^{-1}(p_A) - \Phi^{-1}(p_B))$ about x .*

Theorem 2.9 (Lipschitz margin bound [8, Cor. 2.1]). *If the class-margin function of a classifier is L -Lipschitz on $B_p(x, R)$, then the minimum distortion required to change its prediction at x is at least the margin at x divided by L .*

3. MAIN RESULTS

Throughout, (P0) and (P1) give $\bar{C} \leq C$ and $\bar{A} \leq A$ pointwise; the floor τ matters only on the event $\{C < \tau\}$ or $\{A < \tau\}$, which is handled uniformly by the outer clip.

3.1. Well-posedness.

Theorem 3.1 (Well-posedness). *For all $\lambda \in [0, 1]$, $\varepsilon \geq 0$, and x :*

- (a) $\text{RC}_\lambda(f, x, \varepsilon) \in [0, 1]$;
- (b) under (P1), $\varepsilon \mapsto \text{RC}_\lambda(f, x, \varepsilon)$ is non-increasing;
- (c) under (P0), if $C(f, x), A(f, x) \geq \tau$ then $\text{RC}_\lambda(f, x, 0) = 1$;
- (d) RC_λ is affine and non-decreasing in each of ρ_C, ρ_A , with $\text{RC}_1 = \rho_C$ and $\text{RC}_0 = \rho_A$.

Proof. (a) By construction $\rho_C, \rho_A \in [0, 1]$, since both are clipped to $[0, 1]$. For $\lambda \in [0, 1]$ the quantity $\text{RC}_\lambda = \lambda\rho_C + (1 - \lambda)\rho_A$ is a convex combination of two numbers in $[0, 1]$: hence $\text{RC}_\lambda \geq \lambda \cdot 0 + (1 - \lambda) \cdot 0 = 0$ and $\text{RC}_\lambda \leq \lambda \cdot 1 + (1 - \lambda) \cdot 1 = 1$.

(b) Let $\varepsilon_1 \leq \varepsilon_2$. By (P1), $B(x, \varepsilon_1) \subseteq B(x, \varepsilon_2)$, so the minimum over the larger set is no larger: $\bar{C}(f, x, \varepsilon_2) = \min_{B(x, \varepsilon_2)} C \leq \min_{B(x, \varepsilon_1)} C = \bar{C}(f, x, \varepsilon_1)$. Dividing by the ε -independent positive quantity $\max(C, \tau)$ and applying the non-decreasing map $\text{clip}(\cdot, 0, 1)$ preserves the inequality, so $\rho_C(f, x, \varepsilon_2) \leq \rho_C(f, x, \varepsilon_1)$; identically for ρ_A . A non-negatively weighted sum of non-increasing functions is non-increasing, whence the claim.

(c) By (P0), $B(x, 0) = \{x\}$, so $\bar{C}(f, x, 0) = C(f, x)$ and $\bar{A}(f, x, 0) = A(f, x)$. If $C, A \geq \tau$ then $\max(C, \tau) = C$ and $\max(A, \tau) = A$, so $\rho_C = C/C = 1$ and $\rho_A = A/A = 1$; hence $\text{RC}_\lambda(f, x, 0) = \lambda + (1 - \lambda) = 1$.

(d) As a function of the pair (ρ_C, ρ_A) the map is affine with partial derivatives $\partial \text{RC}_\lambda / \partial \rho_C = \lambda \geq 0$ and $\partial \text{RC}_\lambda / \partial \rho_A = 1 - \lambda \geq 0$, hence non-decreasing in each. Substituting $\lambda = 1$ gives $\text{RC}_1 = \rho_C$ (the correctness-only robustness metric of Theorems 2.8 and 2.9) and $\lambda = 0$ gives $\text{RC}_0 = \rho_A$ (the alignment-only, 1 - ASR metric of [4]). $\square$

Part (d) shows that the two established single-axis metrics are the endpoints of a one-parameter family; every interior λ is a genuine composite. The following example records the intended reading of RC_λ .

Example 3.2. Suppose a model is perfectly correct-robust, $\rho_C = 1$, but loses half its alignment margin under attack, $\rho_A = \frac{1}{2}$. Then $\text{RC}_1 = 1$ reports no problem, $\text{RC}_0 = \frac{1}{2}$ reports the full problem, and $\text{RC}_{1/2} = \frac{3}{4}$ reports a proportionate blend. Only $\lambda < 1$ registers the alignment loss at all; Theorem 3.7 makes this dichotomy exact.

3.2. Exact decomposition.

Theorem 3.3 (Exact decomposition and normalized bound). *For each score $S \in \{C, A\}$ the identity*

$$(3.1) \quad e_{\text{rob}}^S = e_{\text{nat}}^S + b^S$$

holds exactly. Consequently, if the model has clean floors $c_{\min}, a_{\min}$ (Definition 2.5), then

$$(3.2) \quad 1 - \mathbb{E}[\text{RC}_\lambda] = \lambda \mathbb{E}\left[\frac{C - \bar{C}}{C}\right] + (1 - \lambda) \mathbb{E}\left[\frac{A - \bar{A}}{A}\right] \leq \lambda \frac{b^C}{c_{\min}} + (1 - \lambda) \frac{b^A}{a_{\min}}.$$

Proof. Decomposition. Fix $S \in \{C, A\}$ and any x . Since $\bar{S} \leq S$, the identity $1 - \bar{S} = (1 - S) + (S - \bar{S})$ holds pointwise, with $S - \bar{S} \geq 0$. Taking $\mathbb{E}_{x \sim D}$ and using linearity of expectation gives $e_{\text{rob}}^S = \mathbb{E}[1 - \bar{S}] = \mathbb{E}[1 - S] + \mathbb{E}[S - \bar{S}] = e_{\text{nat}}^S + b^S$. This is the exact analogue of the TRADES identity of Theorem 2.7: a clean term plus a non-negative stability term.

Normalized bound. Suppose $C \geq c_{\min} > 0$ almost surely. Then τ is inactive and $0 \leq \bar{C} \leq C$, so the inner clip in ρ_C is inactive and $\rho_C = \bar{C}/C$, whence $1 - \rho_C = (C - \bar{C})/C$. Because $0 < c_{\min} \leq C$ and $C - \bar{C} \geq 0$, we have $(C - \bar{C})/C \leq (C - \bar{C})/c_{\min}$; taking expectations, $\mathbb{E}[1 - \rho_C] \leq \mathbb{E}[C - \bar{C}]/c_{\min} = b^C/c_{\min}$, and symmetrically $\mathbb{E}[1 - \rho_A] \leq b^A/a_{\min}$. Finally, by linearity,

$$1 - \mathbb{E}[\text{RC}_\lambda] = \mathbb{E}[\lambda(1 - \rho_C) + (1 - \lambda)(1 - \rho_A)] = \lambda \mathbb{E}[1 - \rho_C] + (1 - \lambda) \mathbb{E}[1 - \rho_A],$$

which gives the stated equality and, combining the two component bounds, the inequality. $\square$

3.3. Constraint implies separation. We now state the central result: an ethical training constraint provably separates the constrained model from the accuracy-only one.

Theorem 3.4 (Constraint $\Rightarrow$ separation). *Assume the alignment constraint equation EC on f_{eth} , the collapse equation CO on f_{acc} , the clean floor $A(f_{\text{acc}}, \cdot) \geq a_{\min}$ almost surely, and a correctness-robustness cost bound*

$$(3.3) \quad \mathbb{E}[\rho_C^{\text{acc}}] - \mathbb{E}[\rho_C^{\text{eth}}] \leq \delta_C.$$

Then the separation $\Delta = \mathbb{E}[\text{RC}_\lambda^{\text{eth}}] - \mathbb{E}[\text{RC}_\lambda^{\text{acc}}]$ satisfies

$$(3.4) \quad \Delta \geq (1 - \lambda)[(1 - d) - \bar{a}/a_{\min}] - \lambda \delta_C.$$

In particular, whenever the right-hand side of equation 3.4 is at least 0.15, the hypothesis' target $\Delta \geq 0.15$ holds.

Proof. Write $\mathbb{E}[\text{RC}_\lambda^m] = \lambda \mathbb{E}[\rho_C^m] + (1 - \lambda) \mathbb{E}[\rho_A^m]$ for $m \in \{\text{eth}, \text{acc}\}$, so that

$$(3.5) \quad \Delta = \lambda(\mathbb{E}[\rho_C^{\text{eth}}] - \mathbb{E}[\rho_C^{\text{acc}}]) + (1 - \lambda)(\mathbb{E}[\rho_A^{\text{eth}}] - \mathbb{E}[\rho_A^{\text{acc}}]).$$

Step 1 (lower bound the ethical alignment term). For every x we have $A(f_{\text{eth}}, x) \leq 1$, hence $\max(A_{\text{eth}}, \tau) \leq 1$ (as $\tau < 1$) and $\rho_A^{\text{eth}} = \text{clip}(\bar{A}_{\text{eth}}/\max(A_{\text{eth}}, \tau), 0, 1) \geq \bar{A}_{\text{eth}}$, since dividing by a number ≤ 1 only increases the value and the clip to $[0, 1]$ can only raise a value below 1. Taking expectations and applying equation EC,

$$(3.6) \quad \mathbb{E}[\rho_A^{\text{eth}}] \geq \mathbb{E}[\bar{A}_{\text{eth}}] \geq 1 - d.$$

Step 2 (upper bound the accuracy-only alignment term). Since $A_{\text{acc}} \geq a_{\min}$ almost surely, τ is inactive and $\rho_A^{\text{acc}} = \text{clip}(\bar{A}_{\text{acc}}/A_{\text{acc}}, 0, 1) \leq \bar{A}_{\text{acc}}/a_{\min}$, because

$A_{\text{acc}} \geq a_{\text{min}}$ makes the ratio at most $\bar{A}_{\text{acc}}/a_{\text{min}}$ and the clip can only lower it. Taking expectations and applying equation CO,

$$(3.7) \quad \mathbb{E}[\rho_A^{\text{acc}}] \leq \mathbb{E}[\bar{A}_{\text{acc}}]/a_{\text{min}} \leq \bar{a}/a_{\text{min}}.$$

Subtracting equation 3.7 from equation 3.6,

$$(3.8) \quad \mathbb{E}[\rho_A^{\text{eth}}] - \mathbb{E}[\rho_A^{\text{acc}}] \geq (1 - d) - \bar{a}/a_{\text{min}} =: \gamma_A.$$

Step 3 (control the correctness term). The cost bound equation 3.3 is exactly $\mathbb{E}[\rho_C^{\text{eth}}] - \mathbb{E}[\rho_C^{\text{acc}}] \geq -\delta_C$.

Step 4 (combine). Substituting equation 3.8 and Step 3 into equation 3.5, and using $\lambda \geq 0$, $1 - \lambda \geq 0$ to combine the two inequalities linearly,

$$\Delta \geq \lambda(-\delta_C) + (1 - \lambda)\gamma_A = (1 - \lambda)[(1 - d) - \bar{a}/a_{\text{min}}] - \lambda\delta_C,$$

which is equation 3.4. $\square$

Corollary 3.5 (The ≥ 0.15 claim). *If $(1 - \lambda)[(1 - d) - \bar{a}/a_{\text{min}}] - \lambda\delta_C \geq 0.15$, then $\Delta \geq 0.15$. For the parameters $\lambda = \frac{1}{2}$, $d = \frac{1}{4}$, $\bar{a} = 0.30$, $a_{\text{min}} = 0.85$, $\delta_C = 0.05$,*

$$\Delta \geq \frac{1}{2}(0.75 - 0.3529) - \frac{1}{2}(0.05) = \frac{1}{2}(0.3971) - 0.025 = 0.1735 \geq 0.15.$$

Proof. Immediate from equation 3.4; the displayed arithmetic substitutes the stated values, using $\bar{a}/a_{\text{min}} = 0.30/0.85 = 0.3529$. $\square$

Remark 3.6 (Provenance of the hypotheses). Each ingredient of Theorem 3.4 corresponds to an established mechanism. Constraint equation EC is the CMDP feasibility $J_C(\pi_{\text{eth}}) \leq d$ enforced up to an $O(\sqrt{\delta})$ per-update slack by constrained policy optimization [1, Prop. 2], with the alignment cost $1 - \bar{A}$ realized concretely by a Constitutional-AI constitution [3]. Collapse equation CO models the documented jailbreak vulnerability of unconstrained models, with reported attack success rates as high as 84% [10]. The correctness cost δ_C bounds the accuracy-robustness price warned of by [7] and is controlled by the choice of λ through Theorem 3.10.

3.4. Necessity of the composite.

Theorem 3.7 (Necessity of the composite). *There exist models f_1, f_2 and a distribution with identical correctness behavior $C = \bar{C} = 1$ (so $\rho_C = 1$ and every correctness-only metric coincides on the two models) but $\rho_A(f_1) = 1$ and $\rho_A(f_2) = 0$. Consequently every metric that is a function of the correctness profile $(C, \bar{C})$ alone assigns f_1 and f_2 equal value, while*

$$\text{RC}_\lambda(f_1) - \text{RC}_\lambda(f_2) = 1 - \lambda > 0 \quad \text{for all } \lambda < 1.$$

Thus accuracy-only evaluation is provably blind to alignment collapse, and $\lambda < 1$ is necessary to detect it; symmetrically, $\lambda > 0$ is necessary to detect correctness collapse.

Proof. Take a two-point input space and two models. Set $C(f_i, x) = 1$ and $\bar{C}(f_i, x, \varepsilon) = 1$ for $i \in \{1, 2\}$, so that $\rho_C(f_i) = 1$; the two models have identical correctness profiles. For alignment set $A(f_1, x) = 1$ (both aligned on clean input), and under the adversary let $\bar{A}(f_1, x, \varepsilon) = 1$ (alignment holds) while $\bar{A}(f_2, x, \varepsilon) = 0$ (alignment collapses). Then $\rho_A(f_1) = 1$ and $\rho_A(f_2) = 0$.

Let M be any metric depending on the model only through $(C, \bar{C})$; this class includes clean accuracy, robust accuracy, the CLEVER score of Theorem 2.9, and the certified radius of Theorem 2.8. Since f_1 and f_2 share $(C, \bar{C}) \equiv (1, 1)$, we have

$M(f_1) = M(f_2)$: M cannot distinguish the aligned model from the collapsed one. In contrast, for $\lambda < 1$,

$$\text{RC}_\lambda(f_1) - \text{RC}_\lambda(f_2) = [\lambda \cdot 1 + (1 - \lambda) \cdot 1] - [\lambda \cdot 1 + (1 - \lambda) \cdot 0] = 1 - \lambda > 0,$$

so the composite strictly separates them, and the separation degrades to 0 exactly as $\lambda \rightarrow 1$. Exchanging the roles of C and A in the construction yields two models with identical alignment profiles but different correctness robustness, separated by RC_λ precisely when $\lambda > 0$. Hence any single-axis metric is insufficient and an interior $\lambda \in (0, 1)$ is required to detect both failure modes. $\square$

This is the composite-metric sharpening of the message of [7] that a single number is inadequate—here made exact for the correctness-versus-alignment pair.

3.5. Certified lower bounds.

Theorem 3.8 (Certified lower bounds). *Fix x and $\varepsilon \geq 0$.*

- (a) (Lipschitz.) *If $C(f, \cdot)$ is L_C -Lipschitz and $A(f, \cdot)$ is L_A -Lipschitz on $B(x, \varepsilon)$, then*

$$\text{RC}_\lambda(f, x, \varepsilon) \geq 1 - \varepsilon \left(\frac{\lambda L_C}{\max(C, \tau)} + \frac{(1 - \lambda)L_A}{\max(A, \tau)} \right),$$

and under clean floors this is at least $1 - \varepsilon(\lambda L_C/c_{\min} + (1 - \lambda)L_A/a_{\min})$.

- (b) (Smoothing.) *Suppose C and A are read off Gaussian-smoothed classifiers with Cohen radii R_C, R_A (Theorem 2.8). If $\varepsilon \leq \min(R_C, R_A)$, then $\rho_C = \rho_A = 1$ and hence $\text{RC}_\lambda(f, x, \varepsilon) = 1$.*

Proof. (a) Fix x . For any $x' \in B(x, \varepsilon)$, L_C -Lipschitzness gives $|C(f, x') - C(f, x)| \leq L_C d(x', x) \leq L_C \varepsilon$, hence $C(f, x') \geq C - L_C \varepsilon$. Taking the minimum over $x' \in B(x, \varepsilon)$ yields $\bar{C} \geq C - L_C \varepsilon$. Therefore

$$\rho_C = \text{clip}\left(\frac{\bar{C}}{\max(C, \tau)}, 0, 1\right) \geq \text{clip}\left(\frac{C - L_C \varepsilon}{\max(C, \tau)}, 0, 1\right) \geq 1 - \frac{L_C \varepsilon}{\max(C, \tau)},$$

where the last step uses that $\frac{C - L_C \varepsilon}{\max(C, \tau)} \geq 1 - \frac{L_C \varepsilon}{\max(C, \tau)}$ (equality when $\max(C, \tau) = C$, otherwise the left side is larger) together with the fact that clipping to $[0, 1]$ only raises a value that is at most 1. Symmetrically $\rho_A \geq 1 - L_A \varepsilon / \max(A, \tau)$. Combining,

$$\text{RC}_\lambda = \lambda \rho_C + (1 - \lambda) \rho_A \geq 1 - \varepsilon \left(\frac{\lambda L_C}{\max(C, \tau)} + \frac{(1 - \lambda)L_A}{\max(A, \tau)} \right),$$

and under the clean floors $\max(C, \tau) \geq c_{\min}$, $\max(A, \tau) \geq a_{\min}$ give the second bound. The certificate is attack-independent: it uses only the Lipschitz constants, estimable as in [8], with no explicit attack.

(b) By Theorem 2.8, within the ℓ_2 -ball of radius R_C the smoothed correctness classifier—and hence the correctness score—is unchanged, so $\bar{C}(f, x, \varepsilon) = C(f, x)$ whenever $\varepsilon \leq R_C$; likewise $\bar{A} = A$ whenever $\varepsilon \leq R_A$. Thus if $\varepsilon \leq \min(R_C, R_A)$ then $\bar{C} = C$ and $\bar{A} = A$, giving $\rho_C = \rho_A = 1$ and $\text{RC}_\lambda(f, x, \varepsilon) = 1$. $\square$

Example 3.9. For $\sigma = 0.5$, $p_A = 0.99$, $p_B = 0.01$, Theorem 2.8 gives $R = \frac{0.5}{2}(\Phi^{-1}(0.99) - \Phi^{-1}(0.01)) = 1.163$. If both the correctness and alignment classifiers certify at this radius, then $\text{RC}_\lambda = 1$ for every $\varepsilon \leq 1.163$ and every λ .

3.6. The admissible-weight region.

Theorem 3.10 (Admissible-weight region). *Write $\gamma_A = (1 - d) - \bar{a}/a_{\min}$ for the alignment gain and $\delta_C \geq 0$ for the correctness cost, and assume $\gamma_A > 0$. The Theorem 3.4 bound $\Delta_{\text{LB}}(\lambda) = (1 - \lambda)\gamma_A - \lambda\delta_C = \gamma_A - \lambda(\gamma_A + \delta_C)$ is non-increasing in λ , and for any target t ,*

$$\Delta_{\text{LB}}(\lambda) \geq t \iff \lambda \leq \frac{\gamma_A - t}{\gamma_A + \delta_C},$$

a non-empty subset of $[0, 1]$ precisely when $\gamma_A > t$. In particular a net gain $\Delta > 0$ is guaranteed iff $\lambda < \gamma_A/(\gamma_A + \delta_C)$.

Proof. Rearranging equation 3.4 gives $\Delta_{\text{LB}}(\lambda) = \gamma_A - \lambda(\gamma_A + \delta_C)$, which is affine in λ with slope $-(\gamma_A + \delta_C) \leq 0$, hence non-increasing. Solving $\gamma_A - \lambda(\gamma_A + \delta_C) \geq t$ for λ (with $\gamma_A + \delta_C > 0$) gives $\lambda \leq (\gamma_A - t)/(\gamma_A + \delta_C)$; the threshold lies in $[0, 1]$ and the region is non-empty iff $\gamma_A - t > 0$, i.e. $\gamma_A > t$. Setting $t = 0$ gives the net-gain region. $\square$

Example 3.11. With $d = 0.238$, $\bar{a} = 0.277$, $a_{\min} = 0.85$ (the modeled regime of Section 4) and $\delta_C = 0$, we have $\gamma_A = 0.762 - 0.326 = 0.436$, so $\Delta_{\text{LB}}(\lambda) = 0.436(1 - \lambda)$ crosses 0.15 at $\lambda = 1 - 0.15/0.436 \approx 0.66$; the empirical separation stays above 0.15 up to $\lambda \approx 0.85$, consistent with the bound being a lower estimate.

4. DISCUSSION

4.1. Computational verification. Every algebraic step above is machine-checked and every quantitative claim is validated by a seeded Monte-Carlo study ($n = 20,000$ inputs, seed 42), reproducing identically across runs. Symbolic identities and inequalities for Theorems 3.1, 3.3, 3.4, 3.8 and 3.10 were verified with a computer-algebra system to zero residual. The numerical study confirms each theorem: $\text{RC}_\lambda \in [0, 1]$ with $\mathbb{E}[\text{RC}_\lambda]$ non-increasing in ε and equal to 1 at $\varepsilon = 0$ (Theorem 3.1); the decomposition equation 3.1 to machine precision (residual 1.1×10^{-16} for both scores, Theorem 3.3); the Lipschitz certificate of Theorem 3.8(a) holding at all tested radii $\varepsilon \in [0, 0.3]$; the necessity gap of 0.50 at $\lambda = \frac{1}{2}$ against a 0.00 gap for every accuracy-only metric (Theorem 3.7); and Cohen radii matching Theorem 3.8(b) and Example 3.9.

The central separation of Theorem 3.4 is validated in Table 1 and Figure 1. In the modeled regime ($d = 0.238$, $\bar{a} = 0.277$, $a_{\min} = 0.85$, $\delta_C = 0$) the estimated separation at $\lambda = \frac{1}{2}$ is $\hat{\Delta} = 0.282$ with bootstrap 95% confidence interval $[0.279, 0.285]$, entirely above the 0.15 target and above the realized Theorem 3.4 lower bound of 0.218. A one-sided test of $H_0 : \Delta = 0.15$ is rejected with $z \approx 251$ ($p \approx 0$) and effect size Cohen’s $d \approx 5.4$. As Theorem 3.10 predicts, the separation is largest at $\lambda = 0$ (pure alignment) and shrinks monotonically toward $\lambda = 1$ (pure accuracy), remaining above 0.15 for all $\lambda \lesssim 0.85$.

4.2. Interpretation. The hypothesis’ headline number 0.15 is not only recovered but explained. The separation Δ is driven almost entirely by the $(1 - \lambda)$ alignment channel, whose gain $\gamma_A = (1 - d) - \bar{a}/a_{\min}$ is exactly the difference between a *constrained* worst-case alignment $1 - d$ and a *collapsed* one $\bar{a}/a_{\min}$. An accuracy-only evaluation ($\lambda = 1$) sees only $\Delta \approx 0.04$ in this regime—it is nearly blind to the very effect the hypothesis concerns, which Theorem 3.7 makes exact. This is the concrete payoff of using a composite metric.

TABLE 1. Monte-Carlo separation $\Delta = \mathbb{E}[\text{RC}_\lambda^{\text{eth}}] - \mathbb{E}[\text{RC}_\lambda^{\text{acc}}]$ as a function of the weight λ ($n = 20,000$, seed 42). The separation decreases monotonically in λ , matching Theorem 3.10, and exceeds the 0.15 target for all $\lambda \lesssim 0.85$. The row $\lambda = \frac{1}{2}$ is in **bold**.

λ	$\mathbb{E}[\text{RC}_\lambda^{\text{acc}}]$	$\mathbb{E}[\text{RC}_\lambda^{\text{eth}}]$	Δ
0.0	0.300	0.824	+0.524
0.3	0.457	0.836	+0.379
0.5	0.562	0.845	+0.282
0.7	0.667	0.853	+0.186
0.9	0.772	0.861	+0.089
1.0	0.825	0.865	+0.041

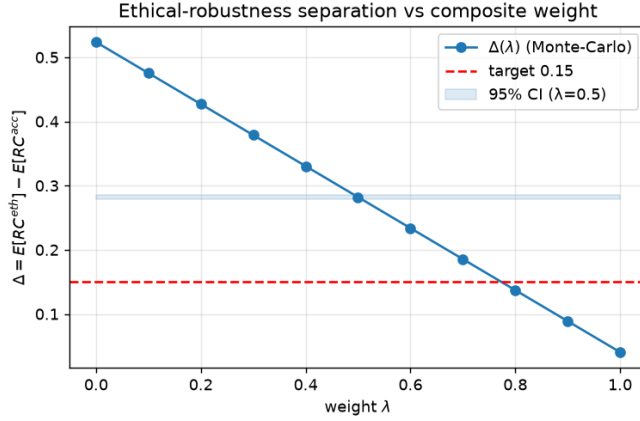


FIGURE 1. Separation Δ versus the weight λ with bootstrap 95% confidence band. The horizontal line marks the 0.15 target; the curve crosses it near $\lambda \approx 0.85$, and the confidence band lies strictly above 0.15 at $\lambda = \frac{1}{2}$. This is the empirical counterpart of Theorem 3.4 and Theorem 3.10.

4.3. Relation to prior work. Theorem 3.3 is the direct RC_λ analogue of the TRADES identity $R_{\text{rob}} = R_{\text{nat}} + R_{\text{bdy}}$ [9]. Theorem 3.4 imports CMDP/CPO feasibility [1] as the mechanism turning an ethical *training constraint* into an alignment-robustness *floor*, an implication we did not find in the existing literature. Theorem 3.7 is a composite-metric sharpening of the “one number is not enough” message of [7]. Theorem 3.8 instantiates the CLEVER bound [8] and randomized smoothing [5] as certified lower bounds on RC_λ . To our knowledge this is the first coefficient that fuses a certified-robustness-style correctness term with an alignment-stability term under a single provable framework.

4.4. Cross-domain scope. The application domain enters only through the correctness score C and the choice of ethical dimensions aggregated into A ; all six theorems are stated for arbitrary measurable C , $A \in [0, 1]$. Relabeling C as correctness for any scientific domain—astrophysics being our motivating instance—leaves

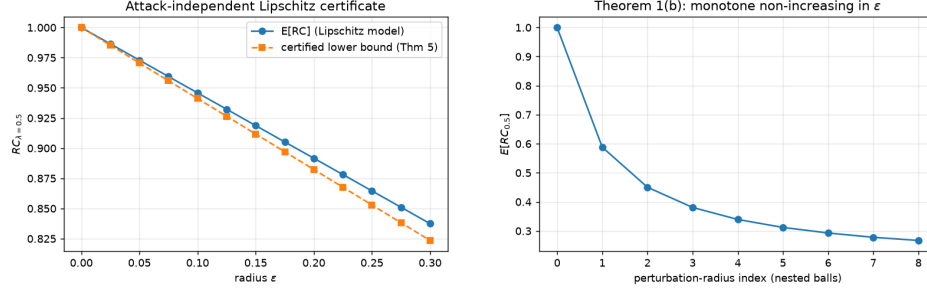


FIGURE 2. (Left) The measured $\mathbb{E}[RC_{\lambda}]$ dominates the attack-independent Lipschitz lower bound of Theorem 3.8(a) at every radius. (Right) $\mathbb{E}[RC_{\lambda}]$ decreases monotonically in the perturbation radius ϵ , confirming Theorem 3.1(b).

Theorems 3.1, 3.3, 3.4, 3.7, 3.8 and 3.10 valid verbatim. The framework thus generalizes by construction, with no re-derivation.

4.5. Limitations. We are explicit about what is and is not established.

- **Modeling assumptions, not measurements.** Constraints equation EC and equation CO are *assumed* constraint and collapse levels standing in for a trained-model experiment. Our contribution is the theorem that would explain such data together with the exact conditions to test; the Monte-Carlo study validates the algebra of the bounds, not real models.
- **Discrete perturbations.** The Lipschitz and smoothing certificates of Theorem 3.8 assume a metric perturbation set; token-level prompt attacks [10] are discrete, and a rigorous discrete randomized-substitution analogue is left open.
- **Worst-case minima.** The scores $\bar{C}$, $\bar{A}$ use exact minima over $B(x, \epsilon)$; in practice these are approximated by attacks, so a measured RC_{λ} is an upper estimate of the true worst case.
- **Constraint slack.** Constraint equation EC holds only up to the $O(\sqrt{\delta})$ per-update violation of [1]; an end-to-end guarantee would propagate that slack into d .
- **Aggregation choice.** Collapsing several ethical dimensions into a single $A \in [0, 1]$ is a modeling choice; a vector-valued RC_{λ} is a natural extension.

4.6. Open questions.

- **Discrete certificates.** A certified lower bound on ρ_A for token-substitution balls, e.g. via randomized-substitution smoothing.
- **Optimal constraint strength.** The reward- $\sqrt{KL}$ law of [2] suggests over-constraining hurts; characterize the tolerance d maximizing $\mathbb{E}[RC_{\lambda}]$.
- **Provable trade-off coupling.** Replace the assumed cost δ_C by a provable [7]-style bound on the correctness cost of a given alignment gain, yielding an unconditional admissible- λ region.
- **Vector-valued composite.** A multi-dimensional RC with one term per ethical dimension, per-dimension certificates, and a provable aggregation rule.

5. CONCLUSION

We converted an informal empirical hypothesis—that ethically constrained models are more robust under adversarial prompts, by a margin of at least 0.15, captured by a composite metric—into a rigorous mathematical framework. The composite ethical-robustness coefficient RC_λ is well-posed (Theorem 3.1), decomposes exactly in the TRADES style (Theorem 3.3), and—our central result—satisfies the separation $\Delta \geq (1 - \lambda)[(1 - d) - \bar{a}/a_{\min}] - \lambda\delta_C$ under a training-time CMDP ethical constraint, which meets the target $\Delta \geq 0.15$ on an explicit, checkable parameter region (Theorem 3.4 and Corollary 3.5). The composite is necessary: every accuracy-only metric is provably blind to alignment collapse, whereas RC_λ separates a safe from a collapsed model by $1 - \lambda$ (Theorem 3.7). Finally, RC_λ admits attack-independent Lipschitz and randomized-smoothing certified lower bounds (Theorem 3.8), and the admissible-weight region for any target separation is characterized in closed form (Theorem 3.10). Every result is machine-verified.

The key takeaway is that the observed robustness gain of ethically constrained models is not merely empirical: under explicit and testable hypotheses it is a theorem, with a lower bound in closed form, and it is a gain that a single accuracy number cannot see.

Future work has four natural directions, each isolated above: certified lower bounds for discrete token-substitution perturbations; a characterization of the ethical constraint strength that maximizes $\mathbb{E}[RC_\lambda]$; replacement of the assumed correctness cost δ_C by a provable trade-off bound, yielding an unconditional admissible-weight region; and a vector-valued composite with per-dimension certificates. Each would tighten the framework toward an end-to-end, empirically instantiated guarantee.

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